21-242 Recitation 12

Span, Linear Independence, Bases — Q5. Every list contains a basis of its span

Course-level solution, with a later-course alternative when useful

October 1, 2026

Prompt

Prove that for any list $\mathcal{L} \subseteq V$, there exists a sublist $\mathcal{L}' \subseteq \mathcal{L}$ such that $\mathcal{L}'$ is a basis for $\text{Span}(\mathcal{L})$.

Course-level solution

Let $\mathcal{L} = (\vec{v}_1, \ldots, \vec{v}_n)$. Build $\mathcal{L}'$ by reading $\mathcal{L}$ from left to right and keeping a vector if and only if it is not in the span of those already kept. (Equivalently: drop each vector that is a linear combination of the preceding vectors of $\mathcal{L}$.)

- $\mathcal{L}'$ is linearly independent by construction (sequential definition).
- $\text{Span}(\mathcal{L}') = \text{Span}(\mathcal{L})$: every dropped vector already lay in the span of earlier kept vectors, so by induction each $\vec{v}_j$ lies in $\text{Span}(\mathcal{L}')$. The reverse inclusion is obvious.

A linearly independent spanning list of $\text{Span}(\mathcal{L})$ is a basis of $\text{Span}(\mathcal{L})$.

Later-course solution (using ideas from the rest of the handout)

Once you have the extension lemma and dim: take a maximal linearly independent sublist $\mathcal{L}' \subseteq \mathcal{L}$. Maximality implies it spans $\text{Span}(\mathcal{L})$, else you could add another vector of $\mathcal{L}$. Finite generation of the span is what makes maximality work without Zorn.